Dhia Gharsallaoui

Senior Backend Engineer - SRE

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Senior Backend Engineer with 4+ years of experience architecting and building high-performance distributed systems. Currently leading backend development at a DeFi trading platform, with deep expertise in Golang, blockchain integrations, and real-time trading infrastructure. Proven track record in crypto startups, building scalable APIs, and implementing mission-critical systems for financial applications.

Technical Skills

Programming Languages

- Go (Golang) - Expert
- Python - Production
- Solidity - Blockchain
- TypeScript
- Scala
- Rust

Backend Architecture

- Microservices Design
- Event-Driven Systems
- Domain-Driven Design
- Clean Architecture
- API Design Patterns
- System Integration

APIs & Protocols

- REST & GraphQL
- gRPC & Protocol Buffers
- WebSocket Real-time
- Message Queues
- Apache Kafka
- NATS Streaming

Databases & Storage

- PostgreSQL
- TimescaleDB
- Redis Caching
- Elasticsearch
- Apache Druid
- Data Modeling

DeFi & Blockchain

- Ethereum Protocol
- Smart Contract Integration
- Cross-chain Bridges
- DEX Trading Systems
- On-chain Data Processing
- Multi-signature Wallets

DevOps & Cloud

- Kubernetes Orchestration
- Docker Containerization
- Terraform IaC
- CI/CD Automation
- Prometheus Monitoring
- Distributed Tracing

Professional Experience

Lead Backend Engineer

Perplex - DeFi Trading Platform • *April 2025 - Present*

Leading backend architecture and development for an on-chain trading platform serving DeFi protocols and traders.

- Architecting and implementing high-performance trading backend systems in Golang, handling real-time price feeds and order execution
- Designing scalable API architecture for DeFi trading operations, supporting multiple blockchain networks and DEX integrations
- Building robust on-chain data ingestion pipelines processing millions of transactions daily with sub-second latency requirements
- Implementing smart contract interaction layers for automated trading strategies and liquidity provision
- Establishing system architecture patterns for high-availability trading infrastructure with zero-downtime deployments
- Collaborating with frontend and protocol teams to deliver end-to-end trading features and risk management systems

Backend Solution Architect

Skilld • *August 2021 - April 2025*

Architected and built scalable backend systems and distributed infrastructure for cloud-native applications.

- Designed and developed high-performance microservices in Golang using domain-driven design principles, serving 10M+ requests daily
- Built event-driven architectures using Apache Kafka and NATS, processing real-time data streams with guaranteed message delivery
- Implemented comprehensive API design patterns with REST and gRPC services, achieving 99.9% uptime across distributed systems
- Developed stream processing pipelines using Apache Spark for real-time analytics and data transformation
- Established monitoring and observability systems with Prometheus, Grafana, and distributed tracing for production debugging
- Implemented database optimization strategies with PostgreSQL and Redis, reducing query response times by 60%
- Built CI/CD pipelines and infrastructure automation reducing deployment time from hours to minutes

Blockchain Backend Developer

ChainsAtlas • *March 2024 - April 2025*

Developed backend systems for blockchain protocols and cross-chain infrastructure at a crypto startup.

- Architected multi-chain bridge backend systems connecting EVM networks with XRPL, handling cross-chain asset transfers
- Built high-performance transaction processing engines with retry mechanisms and comprehensive error handling
- Implemented cryptographic validation systems for multi-signature operations and secure asset management
- Designed blockchain-agnostic APIs enabling protocol deployment across multiple networks
- Developed real-time blockchain monitoring systems with alerting for transaction failures and network issues

Backend Software Engineer

idatase GmbH • *January 2024 - March 2024*

Built scalable data processing backends and APIs for high-volume data ingestion systems.

- Developed high-throughput data ingestion APIs using Python/Flask and PostgreSQL, processing 100K+ records hourly
- Implemented containerized microservices architecture with Docker and Kubernetes deployment
- Built real-time metrics collection systems and alerting infrastructure for monitoring system performance
- Established automated testing and deployment pipelines with GitHub Actions

Education

Master's Degree in Computer Science
MINES ParisTech • *2020 - 2021*

Focus on Distributed Systems and Cloud Computing • Minor in Mathematics

Engineer's Degree in Industrial Engineering
ENIT • *2018 - 2021*

Specialization in Big Data Analytics • Minor in Data Science

Open Source Contributions & Projects

Mage-ai - Data Pipeline Platform
Python • Apache Druid • Authentication Systems

Contributed authentication backend, Apache Druid integration, and logging infrastructure to production data pipeline platform with 8K+ GitHub stars.

Go-mail - Email Client Library
Go • OpenPGP • Middleware Pattern

Architected middleware pattern and implemented OpenPGP encryption for secure email communications, demonstrating advanced Go design patterns.

Go-RTE - API Client SDK
Go • REST API • Client Libraries

Built production-ready Go SDK for RTE APIs with comprehensive error handling, rate limiting, and clean interface design.

Kratos Admin UI - Identity Management
React • TypeScript • Ory Kratos • Identity Systems

Modern, responsive admin interface for Ory Kratos identity management system with identity management, session monitoring, and analytics dashboard.